

When the Evaluator Decides the Result: Measuring Faithfulness in Retrieval-Augmented Generation

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Abstract

Faithfulness in retrieval-augmented generation is usually measured by asking a natural language inference model whether the retrieved context entails the generated answer, aggregated as a maximum over retrieved chunks. We hold a generation grid fixed — four embedding models, two question answering datasets, three generators, 24,000 answers — and vary only the evaluator. Across six dataset-generator cells, the conclusion about whether embedding choice affects faithfulness changes in two of them depending on which evaluator is used, and the cause is measurable in the evaluators rather than in the retrieval systems. Replacing a grounded value in an answer with one absent from the context, the failure of a document question answering system most needs to catch, moves the entailment score by 0.054 on Natural Questions for the most verbose generator against 0.451 for the tersest, and a fixed threshold catches 5-14% of such edits against 57-71%. Answer structure accounts for most of that gap: the score falls as an answer accumulates assertions, and the three generators average 3.08, 1.77 and 1.25 of them. It does not account for all of it, since the generators still differ at a matched single assertion. Including refusals in the evaluation population produces an apparent relationship between retrieval quality and faithfulness whose direction depends on the generator and which disappears once they are excluded. We report what these choices do to a concrete equivalence claim, and the conditions under which such a claim can be stated.

1 Introduction

A document question answering system is given a set of uploaded files and must answer from them. If an invoice records a project budget of \$1,000, an answer stating \$1,500 fails in a specific way: fluent, on topic, citing the right document, wrong in the one place that matters. Such systems are typ-

ically guarded by a faithfulness score, a measure of whether the generated answer is supported by the retrieved text, with answers below a threshold suppressed or flagged.

This paper asks what such a score measures. We do not propose a new evaluator. We take a fixed set of 24,000 generated answers, vary the measurement, and report where the measurement rather than the system decides the conclusion.

Our contributions are:

- A controlled falsification experiment. Replacing one grounded value with a value absent from the retrieved context, everything else held fixed, measures evaluator sensitivity to the error type that matters (Section 5).
- Evidence that the sensitivity depends strongly on answer structure, not only on the evaluator: entailment scores fall as the number of assertions per answer grows, for all three generators tested (Section 5.3).
- A demonstration that the choice of evaluation population — specifically, whether refusals are included — can produce an apparent relationship between retrieval quality and faithfulness where none survives their exclusion (Section 6).
- What an equivalence claim about faithfulness requires: naming the evaluator, reporting the margin as a curve, and anchoring that margin to a measured effect size (Section 8).

The embedding-model grid is a testbed rather than the object of study. It supplies retrieval systems spanning 4.4 and 11.9 NDCG@5 points on the two datasets, with generation and prompting held constant, which is what makes the comparison controlled.

2 Related Work

2.1 Retrieval quality and downstream outcome

Salemi and Zamani (2024) report that human document-level relevance labels, and LLM relevance judgments, correlate only weakly with downstream retrieval-augmented generation performance, and propose eRAG, which labels each retrieved document by the downstream result the system’s own generator produces from it alone. Their strongest baseline is answer-containment labelling, the scheme our qrels use, which retains a moderate correlation (Kendall’s tau 0.31-0.40 on the QA datasets); their correlations decline as k grows and are measured at $k = 50$ against ours at $k = 5$, and their downstream measure is correctness throughout, from a fine-tuned 60M reader. Our starting point is downstream of theirs: we take it as given that retrieval metrics do not directly predict generation quality, and ask which downstream property a measurement instrument can resolve at all — for faithfulness rather than correctness, and for prompted generators, which can abstain.

2.2 Faithfulness measurement

Entailment-based scoring is standard: an inference model is asked whether the retrieved context entails the generated answer. RAGAS (Es et al., 2024) packages this with context and answer relevance metrics, AlignScore (Zha et al., 2023) trains a unified alignment function over several factual-consistency tasks instead of using a general-purpose inference model, and FaithJudge (Tamber et al., 2025) uses a language model as the judge. We do not propose another such metric. We measure how three of them respond to a controlled, known error, and report what that does to conclusions drawn with them.

2.3 Meta-evaluation of faithfulness detectors

A separate line of work evaluates the detectors themselves. TRUE (Honovich et al., 2022), AG-GREFACT (Tang et al., 2023), RAGTruth (Niu et al., 2024) and FaithBench (Bao et al., 2025) assemble annotated corpora and score detectors against them; MiniCheck (Tang et al., 2024) trains compact fact-checkers on unified grounding datasets. Two of their findings bear on ours: no single metric is best in all settings or for all error types, and on summaries where current detectors disagree, detection falls to about 50%.

These compare detectors against human hallucination labels and report detector accuracy. Section 5 is a meta-evaluation too, but with constructed labels: one value is falsified in an otherwise identical answer, so the error is known by construction and every evaluator sees the same minimal pair. Sections 6 onward use no labels at all — we hold the grid, the queries and the answers fixed, swap the evaluator, and ask whether the ranking of the systems under study survives. The two questions come apart: a detector more accurate on a labelled corpus need not resolve differences between the systems a practitioner is choosing between, and it is that second property which decides whether a published comparison describes the systems or the instrument. Xiao et al. (2023) argue that generation metrics should be analysed as measurement instruments with stated reliability and validity.

2.4 Hallucination and grounding analysis

ReDeEP (Sun et al., 2025) localises hallucination to attention heads and feed-forward components. Sinha (2025) reports that embedding-based detection separates synthetic hallucinations well and real ones poorly, the hard cases staying semantically close to a faithful response. Chen et al. (2025) vary and then combine embedding models for mathematics question answering, on the observation that different embedders succeed on different queries. Our analysis is behavioural rather than mechanistic; we do not probe model internals.

3 Definitions

3.1 Faithfulness

For a generated answer a and retrieved context $R = \{d_1, \dots, d_k\}$, the standard aggregate is:

$$F_{\max}(a, R) = \max_{d \in R} \text{NLI}(d \models a) \quad (1)$$

We also evaluate a claim-level aggregate. Writing $C(a)$ for the assertions in a :

$$F_{\text{claim}}(a, R) = \min_{c \in C(a)} \max_{d \in R} \text{NLI}(d \models c) \quad (2)$$

The maximum over chunks is appropriate — any retrieved document may support a claim. The maximum over claims implicit in (1) is not: an answer is only as supported as its least supported assertion, so one unsupported claim should not be masked by several supported ones. Faithfulness

is measured against the generated answer and the retrieved context only; the gold answer is never used. An earlier version of this work differenced NDCG@5 and faithfulness into a gap metric. We retire it: the two terms have no common unit, the difference is negative for every model on Natural Questions because faithfulness (0.85) exceeds retrieval quality (0.79), and any metric of the form $1 - F/RQ$ reproduces the retrieval ranking whenever F is near-constant. Absolute faithfulness is reported throughout, which also keeps visible the case a difference cannot separate, where both quantities are low.

3.2 Correctness and retrieval hit

Correctness is measured against the gold answer and never against the context, so that a grounded wrong answer and an ungrounded right answer remain distinguishable. A retrieval hit means the gold answer string appears in the retrieved context. This is a strong definition of retrieval success, and the Limitations section notes what it does not capture.

4 Experimental Setup

Two datasets are used, Natural Questions (Kwiatkowski et al., 2019) and HotpotQA (Yang et al., 2018), at 1,000 queries each. Four embedding models index each corpus, retrieval takes the top five chunks of 256 tokens with 32-token overlap, and three generators answer from a byte-identical prompt at temperature 0: two proprietary (Claude Haiku 4.5, GPT-4o-mini) and one open-weight run locally (Qwen2.5-7B-Instruct). That gives $4 \times 2 \times 3 \times 1,000 = 24,000$ answers, each re-scored by every evaluator. The four systems — all-mpnet-base-v2 (contrastive), BGE-M3 (multilingual), E5-large-instruct (instruction-tuned) and text-embedding-3-small (proprietary) — reach NDCG@5 of 0.786, 0.794, 0.806 and 0.830 on Natural Questions and 0.705, 0.809, 0.824 and 0.767 on HotpotQA, a spread of 4.4 and 11.9 points, and they do not rank the same way on the two datasets (Appendix A). A single prompt builder enforces parity, one user turn and no system message, since a prompt difference would confound every cross-generator comparison below.

Three evaluators are compared, on identical inputs. NLI-max is (1) with a DeBERTa-v3-large (He et al., 2023) inference model, the common

choice. Claim-min is (2) with the same model and sentence-level claim segmentation, chosen so no second generator enters the measurement loop. AlignScore (Zha et al., 2023) is an independently trained alignment model.

The answers are byte-identical across evaluators. The premises are not, and that difference is part of what the comparison measures, since premise construction follows each evaluator’s intended interface: the entailment scorer sees each of the five chunks separately plus their concatenation truncated to its 512-token window, and takes the maximum; the claim-level scorer applies the same construction per extracted claim; AlignScore takes the concatenated context as one premise and splits it internally. Some of the difference in Section 5 is therefore premise granularity rather than evaluator sensitivity, and Section 5.1 bounds that share. The hypothesis is normalised identically for all three, with markdown emphasis, headings and list markers stripped before scoring. That matters because one generator emits markdown in 76.0% of its attempted answers against 1.3% and 0.3% for the other two, and the entailment model responds to it (Section 7.2).

One refusal rule is used throughout, from a single shared implementation, so every section below describes the same population. An answer counts as a refusal when, after SQuAD-style normalisation, it begins with one of a fixed list of declining phrases. The rule excludes 5,776 of the 24,000 answers (24.1%), from 11.8% on NQ/Qwen to 30.2% on HotpotQA/GPT-4o-mini. All three generators prepend a "Based on the provided context," preamble to everything, refusals included, so the rule strips it before the anchored test; without that step 107 refusals, all Claude’s, would count as attempts.

5 How Sensitive Are These Evaluators to a Known Error?

5.1 A controlled falsification

A faithfulness score is only useful if it responds when an answer stops being faithful, so we construct that condition directly. For each answer containing a numeric value that appears in the retrieved context, we replace the value with one that does not appear there, preserve the surface format, and re-score against the same chunks. Two eligibility rules carry the design: the original value must be present, or the answer was never

grounded, and the replacement must be absent, or the altered answer is accidentally supported and a low score is correct behaviour.

Cases are taken in dataset order, the first 200 eligible answers per system, dataset and generator, so the scored set is a prefix of an already filtered population: 4,784 cases against 7,609 eligible, from a near-census on GPT-4o-mini (800 of 882 on NQ) to two fifths on Claude (800 of 1,946).

Two controls accompany it: an entity substitution, a comparable edit on a named entity rather than a value, and the untouched answer scored against another query’s context, which gives the floor of each evaluator’s scale. Cases are rebuilt deterministically, so all three evaluators see identical inputs.

On Claude’s Natural Questions answers, NLI-max leaves the falsified answer at 0.852 against a floor of 0.474, comfortably above any usable threshold. The same edit on GPT-4o-mini’s answers lands at 0.436, essentially at that generator’s floor of 0.506, and the open-weight generator falls between at 0.574 against a floor of 0.470. Evaluator, context and perturbation code are identical. Only the generator that wrote the answer differs.

We predicted that intermediate position before the arm was run, from its assertion count alone (Section 5.3), and it holds in both datasets under all three evaluators. The evaluator ordering is stable too: AlignScore responds most, claim-min next, NLI-max least, in all six cells. Generators differ in how far apart the evaluators are, not in which responds most.

The evaluators do not share a floor, so detection rates only become comparable once the scale is normalised, and we report two gates. On Claude’s NQ cell an untouched answer scored against an unrelated context averages 0.474 under NLI-max and 0.245 under claim-min, and the distribution is heavy-tailed: its 95th percentile is 0.985 under NLI-max, so an unrelated document clears a fixed 0.5 gate nearly half the time. Our second gate sits at that percentile, where "detected" means the falsified answer is no longer better supported than an unrelated document, which means the same thing on every scale. Read that way NLI-max detects 50.2% of falsifications rather than 5.1%, and its gap to GPT-4o-mini narrows from roughly elevenfold to under twofold. The ordering holds at either gate. The floor-anchored gate is a comparison device rather than an operating point, since it sits above the mean untouched answer.

5.2 The effect is not verbatim copying

A natural explanation is that one generator quotes the retrieved text back, so the evaluator scores a copy of its own premise and one substituted value cannot move it. The data does not support it. Verbatim overlap with the retrieved context is 0.226 for Claude, 0.194 for Qwen and 0.235 for GPT-4o-mini, where the explanation needs overlap to run opposite to sensitivity, and the correlations between overlap and the response to falsification are +0.046, +0.178 and +0.166, where it needs all three to be negative. The separation also survives at matched overlap, in every band including the one above 0.3 (Appendix F).

5.3 Answers with more assertions are scored less sensitively

Segmenting each answer into sentence-level assertions and grouping by count gives a declining relationship in all three generators. It is not strictly monotone: the tail buckets are thin and each generator reverses direction once inside them (Appendix C).

The three generators average 3.08, 1.77 and 1.25 assertions per attempted answer here, with median lengths of 55, 25 and 15 words. That accounts for most of the cross-generator difference in Table 1. A falsified value is a smaller and smaller edit to the hypothesis as correct surrounding text accumulates, and the remainder dominates the entailment judgement. It does not account for all of it: within the single-assertion bucket the generators still differ, +0.102 against +0.409 and +0.485. As a check on the claim-level implementation, every single-assertion answer scores identically under both aggregates ($n = 2,472$, largest absolute difference $1.7e-05$, which is batch-padding noise). That check is what surfaced the normalisation defect in Section 7.2, where it failed on all 131 formatted answers by as much as 0.729.

Claim-level aggregation is a partial remedy. As a fraction of the distance to the random-context floor, the median response on Claude’s answers moves from 0.01 to 0.03, and the minimum moves the right way in 55% of cases; when the falsified assertion is not already the weakest, the minimum does not move at all. A second dilution survives inside assertions, since the three generators still differ at matched assertion count and their single-assertion answers average 23.9, 19.5 and 16.1 words.

Dataset	Evaluator	Claude Haiku 4.5			Qwen2.5-7B			GPT-4o-mini		
		Orig	Fals.	Det.	Orig	Fals.	Det.	Orig	Fals.	Det.
NQ	NLI-max	0.906	0.852	5.1%	0.836	0.574	35.1%	0.887	0.436	56.6%
	Claim-min	0.679	0.514	25.1%	0.693	0.397	48.1%	0.863	0.390	61.0%
	AlignScore	0.866	0.619	24.1%	0.863	0.428	51.8%	0.912	0.272	75.4%
HotpotQA	NLI-max	0.702	0.618	13.9%	0.547	0.277	54.8%	0.577	0.186	70.5%
	Claim-min	0.370	0.255	34.3%	0.424	0.160	66.8%	0.571	0.182	70.6%
	AlignScore	0.786	0.454	48.9%	0.749	0.255	74.5%	0.780	0.119	89.4%

Table 1: Response of three evaluators to a falsified value, on identical inputs, pooled over the four embedding systems ($n = 800$ per cell, 784 on HotpotQA/GPT-4o-mini). "Det." is the share of answers passing a fixed 0.5 threshold before falsification and failing it after. Scale floors, the floor-anchored gate and Wilson 95% intervals on every rate are in Appendix B.

HotpotQA	refusal rate	pooled		answered	mean NLI-max	
		worst-best	spread		spread	attempts
Claude	0.393-0.224	0.085	+0.99	0.018	0.27-0.35	0.69-0.71
Qwen	0.378-0.241	0.053	-0.98	0.003	0.85-0.90	0.57
GPT-4o-mini	0.398-0.232	0.072	-0.92	0.029	0.91-0.94	0.62-0.65

Table 2: Pooling refusals into the faithfulness average gives a column that tracks retrieval quality in every generator ($\rho = \text{Spearman against NDCG@5 over the four systems}$) — in opposite directions. Excluding them collapses the spread and the ordering follows retrieval quality in none of the three.

The third generator makes that residual legible. It sits between the other two in aggregate and does not once assertion count is held fixed: at one assertion it is +0.409, against +0.485 for GPT-4o-mini and +0.102 for Claude. Its intermediate aggregate comes from its assertion *distribution*, not from intermediate per-assertion sensitivity. So the remaining gap belongs to one generator rather than to a smooth verbosity axis, which two generators cannot distinguish and three can.

6 The Evaluation Population Can Manufacture a Relationship

All three generators refuse a substantial share of queries, and they refuse more often when retrieval is worse. On HotpotQA the weakest retriever draws 0.378 to 0.398 refusals and the strongest 0.224 to 0.241. A refusal is not an attempt at a grounded answer, so averaging it in carries a retrieval effect into a faithfulness number without looking as though it does.

The amount carried is large: pooled spreads run 0.053 to 0.085, and excluding refusals leaves 0.003 to 0.029. The direction has nothing to do with refusing. It depends on how the evaluator scores the sentence each generator refuses in. Claude’s refusals score far below its attempts, so

pooling drags the weakest retriever down and the pooled column rises with NDCG@5. Qwen’s and GPT-4o-mini’s score far above theirs, so pooling lifts the weakest retriever and the same column falls. One experiment supports both readings and neither is about faithfulness. The pooled column is close to a monotone function of the refusal rate with a sign set by boilerplate, so anyone defending it cannot fall back on a consistent direction of bias. All faithfulness results here exclude refusals and are paired on the queries every system answered.

6.1 Membership Is Set by a Grader, and Graders Disagree

Excluding refusals means deciding which answers are refusals, and a rule decides it. We compare two: a string match against the refusal templates the prompt offers, and a judge model shown the question, the reference answers and the answer. They agree on most rows, the judge finding four to five more refusals per hundred in every system without reordering them. The judge was run on the two proprietary arms only, so this subsection and the next cover sixteen of the twenty-four cells.

A level shift that preserves the ordering is usually unremarkable. This one is not, because the rows it moves sit at the boundary between refusing and attempting. Swapping detectors moves nineteen rows on NQ/Claude and fifty-five on HotpotQA/Claude, and carries the uncorrected p across 0.05 in two of the eight cells it was run over: NQ/Claude under the entailment aggregate from 0.223 to 0.048, HotpotQA/Claude under AlignScore from 0.151 to 0.049. Both land within a thousandth of the threshold and neither survives correction. The number of cells whose verdict changes between evaluators is the same under either detector, but they are different cells, so we

report the count and treat any single cell as undetermined.

6.2 The Correctness Grader Admits Refusals to the Wrong Population

The same problem returns once faithfulness is conditioned on correctness. String containment marks an answer correct when a reference string occurs in it, and a short reference string can occur inside a refusal; on a 200-answer calibration ten refusals were marked correct this way and the judge over-turned nine. Each such row carries a near-floor score into the correct-answer average and lowers the baseline every correct-versus-incorrect comparison is measured against. Under graded judgement no refusal is correct in any of the sixteen judged cells.

That is enough to change a sign. With containment, and a baseline excluding refusals from the incorrect side only, NQ/Claude answers looked more grounded when wrong than when right by 0.018 to 0.050. Excluding refusals from both sides gives -0.006 to +0.018; grading with the judge gives -0.051 to -0.113. Each step corrects a grader rather than an answer, and all three move the quantity the same way. Tested directly over the sixteen judged cells, a permutation test with Holm correction finds three significant, all negative, and no positive-signed cell with an interval excluding zero. We report no evidence that answers are more grounded when wrong. Intervals are 0.08 to 0.13 wide on 44 to 84 wrong answered rows per cell, so this bears on the general claim rather than on individual systems.

The same accounting applies to the contingency between retrieval success and correctness (Appendix G). The cell where retrieval succeeded and the answer was still wrong bears on whether retrieval quality is sufficient, and 69 to 87 per cent of it was never attempted. Read whole it runs from 17 to 36 per cent of queries and would support the claim that good retrieval often fails to produce a correct answer. Most of what it counts is a system declining to answer, and refusal rates differ by more than twenty points across the systems compared. What is left is real, and 4.5 to 7.9 per cent.

7 The Verdict Depends on the Evaluator

We compare the four retrieval systems on faithfulness, refusals excluded, paired on a common query subset, with a paired bootstrap of 10,000 resam-

ples.

The statistic is the spread, the difference in mean faithfulness between the best and worst of the four systems, with a p-value from a paired bootstrap over that pair re-centred at zero (Dror et al., 2018). It is what a practitioner is exposed to, since they adopt one embedding model and stand to lose the distance to the best alternative. It also summarises two systems chosen for being the extremes, so its p-value is not adjusted for that selection and the intermediate systems never enter it. We treat it as descriptive and give the direction across all four in Section 7.1. An omnibus test would be stricter; we do not run one.

Under the common evaluator no cell of the six shows a detectable difference; under the evaluator Section 5 measures as most responsive to known errors, two do. A result reported only under the first would describe the resolution of the instrument rather than a property of the systems. Both surviving cells are on HotpotQA, which has the wider retrieval spread, and neither is on Claude, whose answers the entailment aggregate resolves least well.

Adding a third generator made the correction stricter, eighteen comparisons rather than twelve, and the count rose anyway. The design does not guarantee that. It also produced the sharpest result here: under AlignScore on HotpotQA all three generators rank the four systems identically (all-mpnet < text-embedding-3-small < BGE-M3 < E5-instruct), while under NLI-max no two agree. Read it carefully. The generators are independent of each other but answer from the *same* retrieved contexts, so this is three generators agreeing about a property of the shared retrieval rather than three replications. What it establishes is that one evaluator resolves that property consistently across generators and another does not, on identical inputs.

The count of two is not invariant to the correction either. Two comparisons sit either side of $\alpha = 0.05$ depending on which are treated as one family. NQ/Claude under AlignScore is $p = 0.0081$ uncorrected, adjusted to 0.122 over the eighteen in the table, 0.032 over the six sharing an evaluator, 0.024 over the three in its own cell. HotpotQA/GPT-4o-mini under claim-min is $p = 0.0070$, adjusted to 0.112, 0.042 and 0.014. Either smaller family makes the count four. Correcting within evaluator is defensible, since the six NLI tests establish which cells are null and so precede the question rather than competing with it. We still

Dataset	Evaluator	Claude Haiku 4.5		Qwen2.5-7B		GPT-4o-mini	
		Spread	p (Holm)	Spread	p (Holm)	Spread	p (Holm)
NQ	NLI-max	0.0107	1.000	0.0055	1.000	0.0130	0.978
	Claim-min	0.0216	0.978	0.0076	1.000	0.0151	0.823
	AlignScore	0.0114	0.122	0.0072	1.000	0.0080	0.978
HotpotQA	NLI-max	0.0232	1.000	0.0123	1.000	0.0267	0.823
	Claim-min	0.0345	0.739	0.0181	1.000	0.0378	0.112
	AlignScore	0.0164	0.823	0.0406	0.0187	0.0461	0.0090

Table 3: The same comparison under three evaluators, Holm-corrected over the eighteen comparisons in the table. Two of six cells change verdict at $\alpha = 0.05$ and survive correction, both under AlignScore and both on HotpotQA. Queries, answers, refusal handling and test are identical throughout. Paired n is 784, 790 and 628 on NQ and 517, 499 and 510 on HotpotQA. With 10,000 resamples the smallest attainable non-zero p is $1e-4$, so the cell at $p = 0.0005$ uncorrected is 5 extreme resamples of 10,000, not a clamped floor.

report the eighteen-comparison family, because it is the most conservative and was fixed before the correction was applied, and choosing a smaller one after seeing that it restores cells is a selection we could not account for. We give the alternative as well as the choice, because a count that moves with the multiplicity family depends on an analysis decision in the same way this paper reports for the choice of evaluator.

7.1 Direction is dataset-dependent

Where a difference exists it does not point the same way. Correlating NDCG@5 with mean faithfulness across the four systems, in all eighteen generator-evaluator combinations (Appendix D), splits the two datasets. Under NLI-max and AlignScore the split is exact: all six HotpotQA coefficients are positive, reaching +1.00 in each AlignScore cell, and all six Natural Questions coefficients are zero or negative, reaching -1.00 for Claude under NLI-max. Better retrieval accompanies more faithful answers on the multi-hop dataset and not on the single-hop one. Composing an answer out of several retrieved pieces plausibly makes grounding sensitive to how much of the evidence is present, where a single-fact answer is not. We do not test that. The experiment varies embedding model rather than manipulating retrieval quality, so none of this is an effect of NDCG@5 itself.

Claim-min breaks the split. Two of its six combinations go the wrong way, both on the open-weight generator and in opposite directions on the two datasets. With two evaluators, or two generators, this table would have shown a clean dataset effect. The third of each is what shows the direction is not a property of the dataset alone.

7.2 A normalisation defect that produced a result

An earlier version of this analysis reported two of the four cells then in the design rather than one. The difference was a change of text normalisation, not of test, data or population: the entailment scorer read the answer as the generator wrote it while the claim-level scorer stripped markdown first, so the two aggregates were not reading the same hypothesis. Markdown is unevenly distributed, appearing in 76.0% of Claude’s attempted answers against 1.3% and 0.3% for the other two, so one generator’s formatting habits were entering a comparison between embedding systems. Stripping it for all three removed the significance of NQ/Claude under AlignScore, the cell Table 3 still places just outside the threshold; neither cell that survives correction is affected, both being on generators that emit almost no markdown.

How it was caught is the transferable part. The defect is invisible in every summary statistic and survives every significance test. What exposed it was an identity the two aggregates must satisfy exactly on single-assertion answers, which failed on all 131 formatted answers then in the grid by as much as 0.729 (Appendix H). Metrics that admit such an identity should be checked against it.

8 What an Equivalence Claim Requires

Four of the six cells in Table 3 show no detectable difference under any evaluator. That is not evidence of equivalence, so we test for it with two one-sided tests (Schuirmann, 1987). TOST turns on the margin, and the margin is a judgement about what difference would matter rather than a fact the data supplies. We sweep it from ± 0.01 to

± 0.10 and report the curve (Appendix E).

The curve is steep. On NQ/Claude under NLI-max no pair is equivalent at ± 0.01 , four of six are at ± 0.02 and all six at ± 0.03 ; on HotpotQA nothing is equivalent below ± 0.05 in any cell. Stating equivalence means choosing a column, and the choice does most of the work.

Section 5 gives a way to choose it. Falsifying one grounded value moves NLI-max by 0.049 to 0.095 on Claude’s answers, so ± 0.05 on that evaluator is about the size of one fabricated fact, and equivalence at that margin says the systems differ by less than one fabrication. The anchor is evaluator-specific and so is the verdict: the same falsification moves claim-min by 0.165 and Align-Score by 0.247 on those answers, and at ± 0.03 on NQ/Claude, NLI-max calls all six pairs equivalent while claim-min calls two. An equivalence claim has to name its evaluator, and say what the margin is worth on that evaluator’s scale.

9 Discussion

For work reporting RAG faithfulness:

- Name the evaluator. "Faithfulness did not differ" is incomplete; two of our six cells reverse between evaluators on identical data.
- Report the evaluation population, and whether membership depends on the system being compared. Pooling refusals here produced a column tracking retrieval quality in all three generators, in opposite directions (Table 2).
- Prefer claim-level aggregation to a maximum over the whole answer, most of all where answers carry several assertions.
- Calibrate the margin against a measured effect before claiming equivalence. A falsification experiment of this kind costs nothing beyond re-scoring.

For a document question answering system, the practical reading of Table 1 is that a grounding threshold on a verbose generator catches a minority of falsified values: 5.1% under the common evaluator, 25.1% under the most responsive one tested. A deterministic check that every value in an answer appears in the retrieved context addresses this directly, recalling every falsified value in all six cells at a false-positive rate of 0.8% to

5.2%. Two qualifications matter. Numbers that assert nothing about the world, such as chunk indices and ordered-list markers, are not expected in the retrieved text, and counting them raises that rate on Claude’s HotpotQA answers to about 39%. The rest are quantities the answer derives rather than copies, which a literal check also flags, so it complements an entailment score rather than replacing one.

Checking symbolic content against the source is not a new proposal and we do not claim it as one (Goodrich et al., 2019; Nan et al., 2021; Zhao et al., 2020), and that numbers are a weak point is documented at the level of the representations too (Wallace et al., 2019). Entailment is separately a loose proxy for attribution: the AIS framework (Rashkin et al., 2023) separates support by an identified source from general plausibility.

What this paper adds is narrower than the check: the controlled minimal pair. One numeric value is replaced in an otherwise byte-identical generation and the same case goes to all three evaluators, so what is measured is each evaluator’s response to a known error on identical input, not its agreement with a human label on some other corpus. That covers numeric values, three generators and two datasets.

10 Conclusion

Holding a 24,000-answer generation grid fixed and varying only the faithfulness evaluator changes the conclusion about whether embedding choice affects faithfulness in two of six dataset-generator cells. The differences between evaluators are directly measurable: a falsified grounded value moves the common entailment aggregate by 0.054 on one generator, 0.262 on a second and 0.451 on a third, and how many assertions an answer contains accounts for most of that gap. A faithfulness result that does not name its evaluator, population, aggregation and equivalence margin is not reproducible in the sense that matters: another group making different standard choices on the same generations would reach a different conclusion.

Limitations

The claim is bounded by the grid it was measured on and by the graders that define its populations.

- Four embedding models, two English datasets and three generators, two propri-

etary and one open-weight. The three span 1.12 to 2.73 assertions per answer, which is the axis the mechanism in Section 5.3 runs along, but they remain three points on it and two of the three are instruction-tuned chat models from the same broad family of training recipes.

- Correctness is graded by a judge model rather than by string overlap. Exact match is 0.000 across all 24,000 answers because the generators do not emit bare answer spans. String containment, the obvious fallback, is not a bound in either direction: against the judge it understates accuracy by 5 to 15 points in twelve cells and overstates it by 3 to 7 points in the four HotpotQA cells under Claude, where long answers mention a reference string while asserting something else. The judge is itself a language model applied to its own outputs, and its agreement with a human annotator on this data has not been established.
- Retrieval hit is defined as the gold answer string appearing in the retrieved context. This is a strong definition, and it does not separate retrieval failure from failures of composition, entity resolution or inference once the evidence is present.
- The Natural Questions sample retains only queries whose annotated answer falls inside the retained context window. This makes the retrieval task more constrained than open-domain RAG, where evidence may be distributed, partial or implicit.
- Pairing on the queries every system answered removes exactly those queries where retrieval quality determined whether an answer was produced. The reported comparison is therefore conditional on that subset, which is narrower than an unconditional comparison.
- Effect sizes are small where detected — 0.046 and 0.041 — against retrieval spreads of 4.4 and 11.9 NDCG@5 points. These are detectable differences, not large ones.
- The three evaluators receive byte-identical answers but not identical premises. The entailment scorer is given each of the five retrieved chunks separately, plus their concatenation truncated to the model’s window, and the maximum is taken; the claim-level scorer applies that construction per extracted claim; AlignScore is given the concatenated context as one premise and performs its own splitting. Each is the evaluator used as its authors intend, but it means part of the difference reported in Section 5 is premise granularity rather than evaluator sensitivity. Rescoring the same cases with the same entailment model on the concatenated premise bounds that share between none and half of the gap depending on the cell, and at about a quarter of it on Claude’s answers. The hypothesis, by contrast, is now normalised identically for all three (Section 7.2); it was not in an earlier version, and that difference alone decided one cell.
- Which answers count as refusals is fixed by a rule rather than given by the data, and that rule sets the evaluation population for every result in Sections V through VIII. It is therefore a researcher degree of freedom as well as a finding: substituting one defensible detector for another moves the rows nearest the boundary between refusing and attempting, and carries the uncorrected p-value across the conventional threshold in two of eight cells (Section 6.1). Those two do not survive correction for multiple comparisons, but the population every reported number is computed on still depends on the choice. The rule used here strips the generators’ standard preamble before matching; without that step 107 refusals, all from one generator, would be graded as attempts.
- The falsification cases are the first 200 eligible answers in dataset order for each system, dataset and generator, rather than a random sample of the eligible answers: 4,784 cases scored against 7,609 eligible across the grid. Eligibility requires a grounded numeric value in the answer together with a replacement value absent from the context, so the scored set is a prefix of an already filtered population. Cells whose eligible count is near 200 are close to a census; the largest cells are sampled at about two fifths.
- The repository configures seven embedding models and this paper reports four:

all-mpnet-base-v2 (contrastive), BGE-M3 (multilingual), E5-large-instruct (instruction-tuned) and text-embedding-3-small (proprietary), chosen to span pretraining paradigms and to include one closed-source arm. GTE-large, Instructor-XL and jina-embeddings-v3 (Sturua et al., 2024) are configured and runnable but were not run at this scale: each additional model is a further generation pass over both datasets and all three generators, and the grid is a testbed for the evaluator comparison rather than a survey of embedding models.

- Correctness is graded only on the two proprietary arms. Section 6.1 and Table 9 therefore cover sixteen of the twenty-four cells, and the open-weight generator enters no result that conditions on correctness. Nothing in Sections 5, 7 or 8 depends on that grading.
- The correct-versus-incorrect faithfulness comparison rests on 44 to 84 attempted wrong answers per cell. Excluding refusals and grading correctness with a judge both reduce that population, and the resulting confidence intervals are 0.08 to 0.13 wide. The tests reported in Section 6.2 bear on the general claim and do not resolve individual systems.

References

- F. S. Bao, M. Li, R. Qu, G. Luo, E. Wan, Y. Tang, W. Fan, M. S. Tamber, S. Kazi, V. Sourabh, M. Qi, R. Tu, C. Xu, M. Gonzales, O. Mendelevitch, and A. Ahmad. 2025. FaithBench: A diverse hallucination benchmark for summarization by modern LLMs. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Short Papers)*, pages 448–461.
- S. Chen, Z. Zhao, and J. Chen. 2025. Confident RAG: Enhancing the performance of LLMs for mathematics question answering through multi-embedding and confidence scoring. *arXiv preprint arXiv:2507.17442*.
- R. Dror, G. Baumer, S. Shlomov, and R. Reichart. 2018. The hitchhiker’s guide to testing statistical significance in natural language processing. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 1383–1392.
- S. Es, J. James, L. Espinosa-Anke, and S. Schockaert. 2024. RAGAS: Automated evaluation of retrieval augmented generation. In *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, pages 150–158.
- B. Goodrich, V. Rao, P. J. Liu, and M. Saleh. 2019. Assessing the factual accuracy of generated text. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 166–175. ArXiv:1905.13322.
- P. He, J. Gao, and W. Chen. 2023. DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing. In *Proceedings of the International Conference on Learning Representations*. ArXiv:2111.09543.
- O. Honovich, R. Aharoni, J. Herzig, H. Taitelbaum, D. Kukliansy, V. Cohen, T. Scialom, I. Szpektor, A. Hassidim, and Y. Matias. 2022. TRUE: Re-evaluating factual consistency evaluation. In *Proceedings of the Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3905–3920. ArXiv:2204.04991.
- T. Kwiatkowski et al. 2019. Natural questions: A benchmark for question answering research. *Transactions of the Association for Computational Linguistics*, 7:453–466.
- F. Nan, R. Nallapati, Z. Wang, C. N. dos Santos, H. Zhu, D. Zhang, K. McKeown, and B. Xiang. 2021. Entity-level factual consistency of abstractive text summarization. In *Proceedings of the Conference of the European Chapter of the Association for Computational Linguistics*, pages 2727–2733. ArXiv:2102.09130.
- C. Niu, Y. Wu, J. Zhu, S. Xu, K. Shum, R. Zhong, J. Song, and T. Zhang. 2024. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 10862–10878.
- H. Rashkin, V. Nikolaev, M. Lamm, L. Aroyo, M. Collins, D. Das, S. Petrov, G. S. Tomar, I. Turc, and D. Reitter. 2023. Measuring attribution in natural language generation models. *Computational Linguistics*, 49(4):777–840. ArXiv:2112.12870.
- A. Salemi and H. Zamani. 2024. Evaluating retrieval quality in retrieval-augmented generation. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 2395–2400. ArXiv:2404.13781.
- D. J. Schuirmann. 1987. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680.

D. Sinha. 2025. The semantic illusion: Certified limits of embedding-based hallucination detection in RAG systems. *arXiv preprint arXiv:2512.15068*.

S. Sturua, I. Mohr, M. K. Akram, M. Gunther, B. Wang, M. Krimmel, F. Wang, G. Mastrapas, A. Koukounas, N. Wang, and H. Xiao. 2024. jina-embeddings-v3: Multilingual embeddings with task LoRA. *arXiv preprint arXiv:2409.10173*.

Z. Sun, X. Zang, K. Zheng, J. Xu, X. Zhang, W. Yu, Y. Song, and H. Li. 2025. ReDeEP: Detecting hallucination in retrieval-augmented generation via mechanistic interpretability. In *Proceedings of the International Conference on Learning Representations*. ArXiv:2410.11414.

M. S. Tamber, S. Kazi, V. Sourabh, and J. Lin. 2025. Benchmarking LLM faithfulness in RAG with evolving leaderboards. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 799–811. ArXiv:2505.04847.

L. Tang, T. Goyal, A. R. Fabbri, P. Laban, J. Xu, S. Yavuz, W. Kryscinski, J. F. Rousseau, and G. Durrett. 2023. Understanding factual errors in summarization: Errors, summarizers, datasets, error detectors. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 11626–11644.

L. Tang, P. Laban, and G. Durrett. 2024. MiniCheck: Efficient fact-checking of LLMs on grounding documents. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 8818–8847. ArXiv:2404.10774.

E. Wallace, Y. Wang, S. Li, S. Singh, and M. Gardner. 2019. Do NLP models know numbers? probing numeracy in embeddings. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing and the International Joint Conference on Natural Language Processing*, pages 5307–5315. ArXiv:1909.07940.

Z. Xiao, S. Zhang, V. Lai, and Q. V. Liao. 2023. Evaluating evaluation metrics: A framework for analyzing NLG evaluation metrics using measurement theory. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 10967–10982.

Z. Yang, P. Qi, S. Zhang, Y. Bengio, W. W. Cohen, R. Salakhutdinov, and C. D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Y. Zha, Y. Yang, R. Li, and Z. Yu. 2023. AlignScore: Evaluating factual consistency with a unified alignment function. In *Proceedings of the Annual Meeting of the Association for Computational Linguistics*, pages 11328–11348.

Embedding model	Paradigm	NDCG@5 (NQ)	NDCG@5 (HotpotQA)
all-mpnet-base-v2	contrastive	0.786	0.705
BGE-M3	multilingual	0.794	0.809
E5-large-instruct	instruction-tuned	0.806	0.824
text-embedding-3-small	proprietary	0.830	0.767
spread		4.4 pts	11.9 pts

Table 4: Retrieval quality of the four systems in the testbed. The ranking differs between datasets, and the spread is far larger on HotpotQA.

Z. Zhao, S. B. Cohen, and B. Webber. 2020. Reducing quantity hallucinations in abstractive summarization. In *Findings of the Association for Computational Linguistics: EMNLP*, pages 2237–2249. ArXiv:2009.13312.

A Retrieval quality of the four systems

B Falsification response in full

C Response to falsification by assertion count

D Correlation between retrieval quality and faithfulness

Spearman correlation between NDCG@5 and mean faithfulness over the four retrieval systems, in every generator-evaluator combination. With four systems these coefficients carry no significance individually; the pattern across them is what Section 7.1 reads. The one non-canonical value, +0.63, arises from a tie between two systems.

E Equivalence across margins

Two one-sided tests on per-query faithfulness, over the six pairs of retrieval systems in each cell, across five margins. A claim of equivalence is a claim about one column of this table and one row of it.

F Verbatim overlap and the response to falsification

Overlap is the fraction of answer word 5-grams occurring in the retrieved context as a contiguous token sequence, on tokens with punctuation mapped to spaces, so it is unaffected by the markdown normalisation of Section 7.2. Answers under five tokens are excluded rather than scored zero, since one generator writes far more of them and scoring them zero would manufacture the difference the control tests for. Grouping the 4,784 cases into four overlap bands leaves the mean response to falsification at +0.052 to +0.085 for Claude, +0.202 to +0.364 for Qwen and +0.326 to +0.507 for GPT-4o-mini. The ordering is the same in all four bands,

Dataset / generator	Evaluator	Orig	Falsified	Floor	Detected @0.5	Detected @floor gate
NQ / Claude Haiku 4.5	NLI-max	0.906	0.852	0.474	5.1% [3.8-7.0]	50.2% [45.7-54.7]
	Claim-min	0.679	0.514	0.245	25.1% [21.7-28.9]	58.3% [53.3-63.1]
	AlignScore	0.866	0.619	0.331	24.1% [21.3-27.3]	77.9% [74.4-81.1]
NQ / Qwen2.5-7B	NLI-max	0.836	0.574	0.470	35.1% [31.7-38.8]	71.0% [66.1-75.4]
	Claim-min	0.693	0.397	0.378	48.1% [44.0-52.2]	79.0% [74.6-82.8]
	AlignScore	0.863	0.428	0.240	51.8% [48.2-55.4]	75.5% [72.1-78.7]
NQ / GPT-4o-mini	NLI-max	0.887	0.436	0.506	56.6% [52.9-60.1]	85.2% [81.9-88.1]
	Claim-min	0.863	0.390	0.490	61.0% [57.4-64.6]	91.0% [88.1-93.2]
	AlignScore	0.912	0.272	0.160	75.4% [72.2-78.4]	82.1% [79.2-84.7]
HotpotQA / Claude	NLI-max	0.702	0.618	0.427	13.9% [11.3-17.0]	49.7% [42.3-57.2]
	Claim-min	0.370	0.255	0.161	34.3% [29.0-40.0]	48.9% [41.8-56.1]
	AlignScore	0.786	0.454	0.475	48.9% [45.2-52.5]	93.7% [90.7-95.8]
HotpotQA / Qwen2.5-7B	NLI-max	0.547	0.277	0.245	54.8% [50.1-59.5]	69.6% [64.0-74.7]
	Claim-min	0.424	0.160	0.178	66.8% [61.6-71.6]	76.9% [71.3-81.7]
	AlignScore	0.749	0.255	0.304	74.5% [71.1-77.7]	96.6% [94.4-98.0]
HotpotQA / GPT-4o-mini	NLI-max	0.577	0.186	0.201	70.5% [66.2-74.6]	83.8% [79.7-87.2]
	Claim-min	0.571	0.182	0.204	70.6% [66.2-74.6]	84.2% [80.1-87.6]
	AlignScore	0.780	0.119	0.234	89.4% [86.8-91.6]	97.5% [95.8-98.5]

Table 5: Table 1 with the scale floors, the floor-anchored gate and Wilson 95% intervals. "Floor" is the mean score of an untouched answer against an unrelated query’s context; the floor gate is the 95th percentile of that distribution, per cell and evaluator.

Assertions	Claude Haiku 4.5			Qwen2.5-7B			GPT-4o-mini		
	n	NLI-max	Claim-min	n	NLI-max	Claim-min	n	NLI-max	Claim-min
1	91	+0.102	+0.102	413	+0.409	+0.409	680	+0.485	+0.485
2	300	+0.076	+0.151	205	+0.152	+0.219	77	+0.360	+0.467
3	194	+0.032	+0.229	103	+0.051	+0.150	26	+0.041	+0.401
4	103	+0.024	+0.162	40	+0.024	+0.076	10	+0.204	+0.192
5 or more	112	+0.024	+0.146	39	+0.072	+0.126	7	+0.032	+0.002

Table 6: Drop in score when one grounded value is falsified, by number of assertions in the answer, on Natural Questions. Under NLI-max the response decays as answers accumulate assertions; under claim-level aggregation it does not. The GPT-4o-mini rows beyond two assertions rest on 26, 10 and 7 cases and should be read as such. In the single-assertion row the two aggregates coincide exactly, which is the degenerate case they must reduce to.

including the band above 0.3 where all three generators reuse a substantial part of the retrieved wording, and Claude is separated from GPT-4o-mini by at least fourfold throughout. The 95% confidence intervals on the three correlations are -0.004 to +0.096, +0.127 to +0.229 and +0.118 to +0.212.

G Retrieval success against correctness

H The markdown normalisation defect

Sizing the defect before the fix, on 300 answers per cell, stripping markdown lowers Claude’s whole-answer entailment score by 0.005 to 0.012 on Natural Questions and 0.021 to 0.043 on HotpotQA, moving it by more than 0.05 on 4.7 to 9.7% and 17.7 to 19.3% of answers respectively. The single-assertion identity is the diagnostic: with one claim and no markdown the whole-answer and claim-level aggregates must be numerically equal, and they were, to 1.7e-05. With markdown they dif-

fered on every one of the 131 formatted answers then in the grid, by up to 0.729 on a pair of asterisks. After stripping in both places the formatted group behaves like the unformatted one: 7 of 140 cases differ at all, by at most 1.5e-05, which is batch-padding noise. Over the present grid the check covers 2,472 single-assertion answers with a largest absolute difference of 1.66e-05.

Generator	Evaluator	Natural Questions	HotpotQA
Claude	NLI-max	-1.00	+0.20
	Claim-min	-0.60	+0.60
	AlignScore	-0.20	+1.00
Qwen	NLI-max	-0.80	+0.60
	Claim-min	+0.63	-0.40
	AlignScore	-0.40	+1.00
GPT-4o-mini	NLI-max	-0.80	+0.80
	Claim-min	-0.80	+0.80
	AlignScore	0.00	+1.00

Table 7: With four systems these coefficients cannot carry significance individually. Under NLI-max and AlignScore the pattern is clean in all twelve combinations: every HotpotQA coefficient is positive and every Natural Questions coefficient is zero or negative. Under claim-min it fails in two of six, both on the open-weight generator. The one non-canonical value, +0.63, arises from a tie between two systems.

Dataset / generator	Evaluator	± 0.01	± 0.02	± 0.03	± 0.05	± 0.10
NQ / Claude	NLI-max	0/6	4/6	6/6	6/6	6/6
	Claim-min	0/6	0/6	2/6	6/6	6/6
	AlignScore	3/6	6/6	6/6	6/6	6/6
NQ / Qwen	NLI-max	0/6	3/6	6/6	6/6	6/6
	Claim-min	0/6	0/6	6/6	6/6	6/6
	AlignScore	1/6	6/6	6/6	6/6	6/6
NQ / GPT-4o-mini	NLI-max	0/6	3/6	6/6	6/6	6/6
	Claim-min	0/6	3/6	6/6	6/6	6/6
	AlignScore	1/6	6/6	6/6	6/6	6/6
HotpotQA / Claude	NLI-max	0/6	0/6	0/6	5/6	6/6
	Claim-min	0/6	0/6	0/6	3/6	6/6
	AlignScore	0/6	2/6	5/6	6/6	6/6
HotpotQA / Qwen	NLI-max	0/6	0/6	0/6	6/6	6/6
	Claim-min	0/6	0/6	0/6	6/6	6/6
	AlignScore	0/6	0/6	0/6	4/6	6/6
HotpotQA / GPT-4o-mini	NLI-max	0/6	0/6	1/6	4/6	6/6
	Claim-min	0/6	0/6	1/6	4/6	6/6
	AlignScore	0/6	0/6	0/6	4/6	6/6

Table 8: System pairs judged equivalent, of six tested, across margins. A claim of equivalence is a claim about a particular column — and about a particular row: at ± 0.03 on NQ/Claude, NLI-max calls all six pairs equivalent and claim-min two, on the same answers.

Retrieval hit and answer incorrect	Claude, NQ	Claude, HQA	GPT, NQ	GPT, HQA
Share of all queries	17.4%	34.0%	31.7%	36.3%
Of that cell, refusals	69.3%	86.8%	79.0%	78.1%
Attempted and wrong	5.4%	4.5%	6.7%	7.9%

Table 9: Correctness from graded judgement. The raw cell is 17 to 36 per cent of queries; the share that is a grounded, committed, wrong answer is 4.5 to 7.9 per cent.